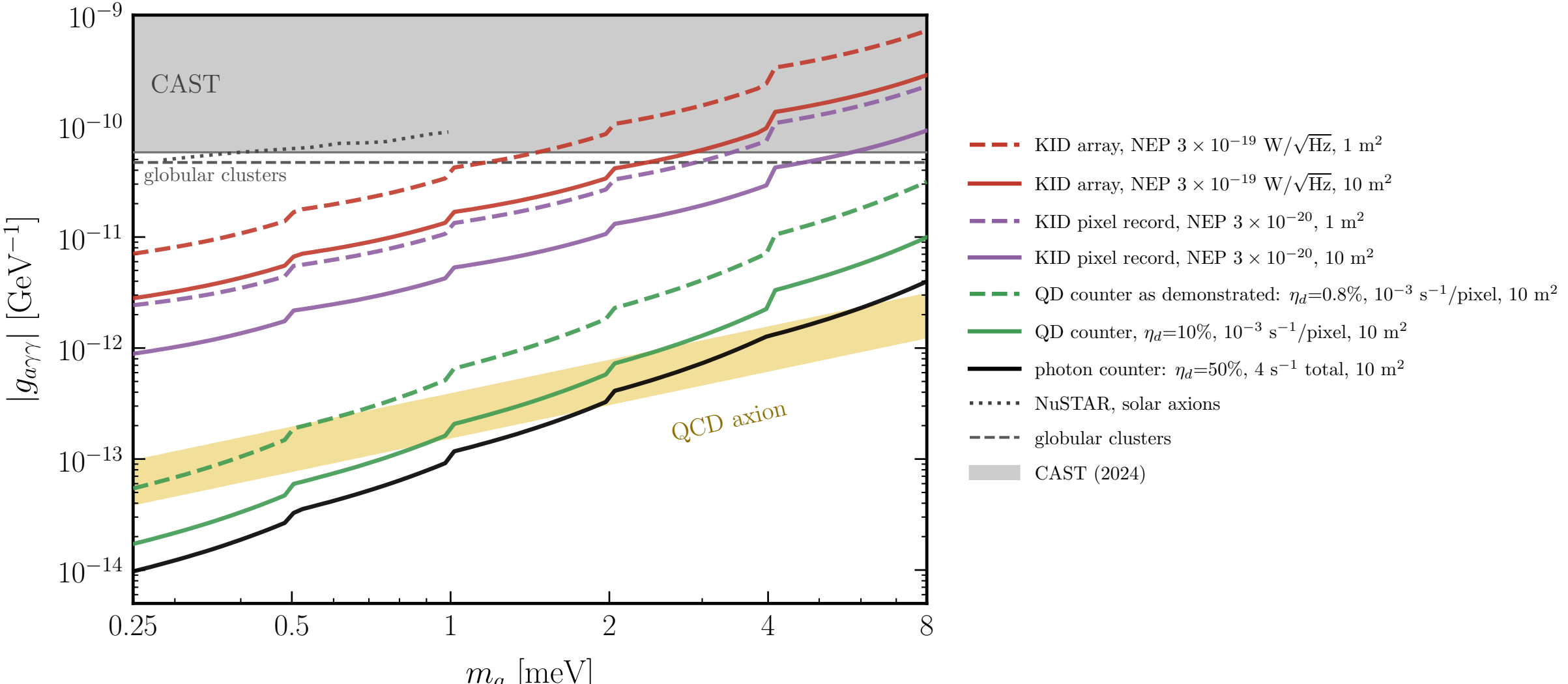

CAST
globular clusters
QCD axion
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
$|g_{a\gamma\gamma}|$ [GeV^{-1}]
0.25
0.5
1
2
4
8
m_a [meV]
KID array, NEP 3×10^{-19} W/$\sqrt{\text{Hz}}$, 1 m^2
KID array, NEP 3×10^{-19} W/$\sqrt{\text{Hz}}$, 10 m^2
KID pixel record, NEP 3×10^{-20}, 1 m^2
KID pixel record, NEP 3×10^{-20}, 10 m^2
QD counter as demonstrated: η_d=0.8%, 10^{-3} s^{-1}/pixel, 10 m^2
QD counter, η_d=10%, 10^{-3} s^{-1}/pixel, 10 m^2
photon counter: η_d=50%, 4 s^{-1} total, 10 m^2
NuSTAR, solar axions
globular clusters
CAST (2024)